

Structured bilevel optimization with many follower variables: Global responses, accuracy, and structural boundaries

1 Problem setting and foundations

Many leader–follower models contain a large number of local follower choices but only a few shared resources, aggregate performance measures, and leader decisions. A tariff may influence the operation of many units, while their nonlinear interaction depends only on total load or emissions. The relevant question is whether the number of shared quantities can control the complexity of global bilevel optimization even as the number of follower variables grows. The distinction between the dimension of the follower and the dimension needed to describe its responses is central.

A tractable follower problem alone does not answer this question. The leader optimizes over the responses induced by all its possible decisions; this optimization can be nonconvex even if every follower has a unique convex minimizer. Conversely, nonconvexity confined to a fixed number of aggregate coordinates need not prevent an exact description of all global responses. The development below separates three tasks: describing global follower responses, optimizing over those responses under a specified selection convention, and representing the requested output at a stated accuracy.

1.1 Related work and the structural question

Classical positive results for optimistic linear bilevel optimization fix the number of *follower* variables. Liu and Spencer [17] give a polynomial algorithm under that restriction; Deng [9] discusses the resulting complexity theory. This restriction differs from fixing the number of leader variables while allowing the follower dimension to grow. Indeed, Sugishita and Carvalho [19] prove NP-completeness with one leader variable, unit-bounded variables, and no additional upper constraints for a linear bilevel class with coupled follower constraints. Their polynomial algorithm for finding a local optimum does not give global tractability.

Ketkov and Prokopyev [14] distinguish fixed follower variable and constraint counts, optimistic and pessimistic responses, and linear and quadratic models. Their optimistic convex-quadratic positive result fixes the follower dimension and imposes their stated convex quadratic upper structure. Neither that result nor a restriction on the total follower constraint count encompasses a model with arbitrarily many local follower variables and local bounds, few shared constraints, and a possibly nonconvex polynomial aggregate cost. In particular, counting shared resource rows separately from local rows is essential. Their pessimistic convention also illustrates why upper feasibility must be specified before complexity statements are compared.

The local elimination tools have direct antecedents. Megiddo and Tamir [18] use low-dimensional multiplier arrangements for separable convex quadratic optimization and for fixed-dimensional quadratic blocks with shared constraints. The local quadratic formulas and multiplier partitions used here build on this principle. Exact sign determination and quantifier elimination in fixed dimension are classical real algebraic algorithms [3]. The structural question is how these ingredients

provide a description of *globally* optimal responses when the aggregate cost is nonconvex, and how that description supports the upper optimization. Stationarity alone cannot supply this last step.

For polynomial local costs, earlier work already supplies relevant approximation methods. Hochbaum and Shanthikumar [12] give logarithmic accuracy dependence for continuous separable resource allocation under their function-oracle and constraint-matrix assumptions. Vigneron [20] gives fully polynomial approximation schemes for global optimization of sums of nonnegative algebraic functions of constant description complexity, with polynomial dependence on inverse relative error. A high-accuracy follower evaluation at a fixed leader is different from a global approximation algorithm for the induced, possibly signed upper objective. Both the numerical degree and the accuracy dependence must be tracked in making that comparison.

Cost-near-optimal follower behavior is an established form of decision uncertainty. Besançon, Anjos, and Brotcorne [6] study robustness of upper feasibility against such responses; Beck, Ljubić, and Schmidt [4] place this model in the broader uncertainty literature. The robust model below also allows a worst-case upper objective. The structural contribution concerns exact compression of its adversarial problems, including nonconvex nominal followers, rather than the introduction of near-optimality robustness.

Finally, polynomial-size optimality certificates need not produce polynomial-time optimization. Buchheim [7] proves NP membership for linear bilevel optimization and polynomial encoding bounds for valid big- M values in optimistic KKT reformulations. Practical reformulations, branching, cuts, and decomposition are surveyed by Kleinert et al. [15]. They provide the computational context for the specializations studied here. Polynomial time for fixed structural dimensions is a worst-case statement; it is not by itself a claim of favorable numerical performance.

1.2 The quadratic-block model

Every polynomial datum in this model is supplied explicitly with rational coefficients.

The leader chooses $x \in C \subset \mathbb{R}^r$, where C is a compact set specified by a rational quantifier-free polynomial formula. The follower vector is $z = (z_1, \dots, z_B)$, with $z_b \in \mathbb{R}^{d_b}$, $d_b \leq d$, and total dimension $N = \sum_{b=1}^B d_b$. Both B and N may grow. For the primary model, let

$$P_b(x) = \{y \in \mathbb{R}^{d_b} : E_b y = e_b(x), G_b y \leq h_b(x)\}, \quad (1)$$

$$P(x) = \{z : z_b \in P_b(x) \ (b = 1, \dots, B), Az = b(x), Hz \leq a(x)\}. \quad (2)$$

The matrices E_b, G_b, A, H are rational and independent of x . There are k_+ shared equality rows and k_- shared inequality rows, and $k = k_+ + k_-$. The number of local rows is unrestricted. All right-hand sides are polynomial in x . Among the local inequalities are coordinate bounds $\ell_{bi}(x) \leq z_{bi} \leq u_{bi}(x)$, with polynomial endpoints satisfying $\ell_{bi}(x) \leq u_{bi}(x)$ on C . Thus the union of the follower feasible sets is bounded. A set $P_b(x)$ or $P(x)$ may be empty; follower feasibility is not a promise at every leader.

Let $U(x) \in \mathbb{Q}[x]^{s \times N}$, and partition its columns into blocks $U_b(x)$. For rational polynomial vectors $c_b(x) \in \mathbb{Q}[x]^{d_b}$, the follower objective is

$$f(x, z) = \sum_{b=1}^B \left(\frac{1}{2} z_b^\top Q_b(x) z_b + c_b(x)^\top z_b \right) + \phi(x, U(x)z). \quad (3)$$

Each $Q_b(x)$ is a symmetric polynomial matrix positive definite for every $x \in C$. The polynomial $\phi(x, w)$ can be nonconvex in w . Its argument $w = U(x)z \in \mathbb{R}^s$ collects the shared aggregate quantities. Positive definiteness is imposed on the local blocks, not on the entire follower Hessian. In particular, the follower need not have a unique minimizer.

The upper objective $F(x, z)$ and upper rows $g_j(x, z) \leq 0$ ($j = 1, \dots, m$) are explicitly encoded rational polynomials. Their monomials may involve many different follower coordinates; neither m nor their support sizes are structural parameters. Unless a theorem says otherwise, the fixed parameters of the exact quadratic model are

$$(r, d, s, k_+, k_-). \quad (4)$$

The scalar local model has $d = 1$ and only interval constraints within a block. Extensions with polynomially varying constraint normals will be stated separately. The approximation results use strictly convex univariate polynomial local costs and a convex aggregate cost, with their own leader-domain and data assumptions; those assumptions are not implicit extensions of (3).

1.3 Follower selection and upper feasibility

For $P(x) \neq \emptyset$, compactness gives the value and solution set

$$v(x) = \min_{z \in P(x)} f(x, z), \quad S(x) = \{z \in P(x) : f(x, z) = v(x)\}. \quad (5)$$

Put $S(x) = \emptyset$ if $P(x) = \emptyset$, and use $v(x)$ only on feasible leaders. The follower always optimizes over $P(x)$ without upper constraints. Define the optimistic feasible set and value by

$$\mathcal{O} = \{(x, z) : x \in C, z \in S(x), g_j(x, z) \leq 0 \ (j = 1, \dots, m)\}, \quad (6)$$

$$\text{val}_{\text{opt}} = \inf_{(x, z) \in \mathcal{O}} F(x, z). \quad (7)$$

Thus the leader may select a favorable global response that satisfies its upper rows. A minimizer is asserted only when attainment is established.

For pessimistic optimization, upper feasibility is universal:

$$D_0 = \{x \in C : S(x) \neq \emptyset, g_j(x, z) \leq 0 \text{ for all } z \in S(x), j = 1, \dots, m\}, \quad (8)$$

$$W_0(x) = \max_{z \in S(x)} F(x, z), \quad \text{val}_{\text{pes}} = \inf_{x \in D_0} W_0(x). \quad (9)$$

The maximum at a fixed feasible leader is attained. The infimum over leaders need not be. An upper constraint cannot discard an inconvenient follower optimum before W_0 is computed.

Given a nonnegative polynomial budget $\rho(x)$ on C , the set of cost-near-optimal responses is

$$S_\rho(x) = \{z \in P(x) : f(x, z) \leq v(x) + \rho(x)\} \quad (10)$$

at feasible leaders, and is empty otherwise. Replacing $S(x)$ by $S_\rho(x)$ in (8)–(9) defines D_ρ , W_ρ , and val_ρ . A bounded budget may itself be a leader coordinate when the leader model permits it. Near-optimal points need not satisfy follower stationarity. The robust compression theorem will restrict the number of affine measurements of z used by each upper criterion; this is an additional hypothesis, distinct from the unrestricted polynomial upper data in the exact optimistic model.

For revenue maximization, the corresponding optimistic objective is a supremum and the pessimistic objective maximizes the minimum revenue over all responses. Equivalently, negate revenue in the minimization conventions above. The exact algorithms report infeasibility separately and state whether the finite optimum value is attained. Approximation algorithms have the guarantees stated in their respective theorems: an original-infeasibility certificate, output satisfying relaxed upper rows, and an empty inner surrogate have different meanings. In particular, neither relaxed output nor an empty inner surrogate decides original feasibility in general.

1.4 Input length, degree, and output

The bit model counts arithmetic on binary-encoded rational data. Let L be the total rational input encoding length, including dimensions, row indices, monomial lists, exponents, and the Boolean formula for C . Let $\delta \geq 2$ bound the actual total degree of every input polynomial. The exact algorithms have running time polynomial in (L, δ) for fixed (4). A claim of polynomial time in L alone assumes that δ is polynomially bounded by L , as for dense or unary-exponent encodings. A sparse monomial representation with unrestricted binary exponents does not have that property. Polynomial local costs in the approximation results are supplied densely; a power-specific result will state its dependence on the numerical power.

A common-field algebraic output consists of a square-free polynomial $p \in \mathbb{Z}[T]$, a rational isolating interval selecting a real root α , and rational polynomials p_i, q_i such that each returned coordinate equals $p_i(\alpha)/q_i(\alpha)$ with $q_i(\alpha) \neq 0$. The guarantee includes the degree of p , all coefficient bit lengths, and the combined encoding of all coordinates. They are bounded polynomially in (L, δ) in the stated fixed dimensions, and hence polynomially in L under the degree-encoding condition above. The polynomial p need not be a minimal polynomial. It suffices that its chosen root specifies the same real extension for every recovered coordinate. If several adversarial problems are requested independently, a common-field guarantee for each leader-witness pair does not imply a polynomial-degree field containing all their unrelated witnesses at once.

An accuracy-bit guarantee takes a positive rational tolerance ε , whose encoding contributes to the running time. Writing $\varepsilon = 2^{-\beta}$ illustrates the distinction between a bound polynomial in β and one polynomial in $1/\varepsilon$. In the base approximation model, the output is a rational leader that is exactly feasible for the base leader domain and admits a feasible follower, with an additive guarantee for the true induced upper objective. An approximate follower vector does not replace the true response in that guarantee. Exact satisfaction of response-dependent upper rows requires additional conditions; without them, a theorem must specify its allowed upper violation. Exact algebraic output, rational near-optimal output, and numerical evaluation are different computational tasks.

The exponents of the polynomial bounds depend on the structural parameters. For the decision versions this gives XP-type bounds, $L^{h(r,d,s,k)}$, under the degree convention. It does not establish a fixed-parameter bound $h(r,d,s,k)L^c$ with an absolute exponent c .

1.5 Algebraic and polyhedral tools

We use the following standard consequences of fixed-dimensional real algebraic computation [3, Theorem 1.3.1 and Section 3.1.3]. For a rational polynomial family in a fixed total number of real variables, realizable sign conditions, including zero signs, can be enumerated in time polynomial in the number of polynomials, their degrees, and coefficient bit lengths. A first-order formula with a fixed total number of free and bound variables can be replaced by an equivalent quantifier-free formula within the same kind of bound. Nonempty semialgebraic sets admit algebraic samples of polynomial degree and encoding length. Applied simultaneously to all free coordinates, sampling gives the common-field representation described above. The total number of variable occurrences introduced by quantified copies must be controlled; fixing only the number of free variables is insufficient.

One elementary encoding fact will be used repeatedly.

Lemma 1.1 (Substitution in fixed dimension). *Let $v \in \mathbb{R}^q$, with q fixed, and suppose that $Z_1(v), \dots, Z_N(v)$ and $D(v)$ have explicit polynomial encodings of polynomial size and degree. On a region where $D > 0$, put $z_i = Z_i/D$. Let $p(x, z)$ be an explicitly listed rational polynomial of total degree at most δ , with the coordinates of x among those of v . Then*

$$D(v)^\delta p(x, Z(v)/D(v))$$

is a polynomial that can be constructed in time polynomial in the input encoding lengths, the degrees of Z_i, D , and δ . Its sign agrees with that of $p(x, z)$ on the region.

Proof. An input monomial $ax^\nu z^\gamma$ becomes $ax^\nu \prod_i Z_i^{\gamma_i} D^{-|\gamma|}$. Its degree is at most $\delta(1+T)$ if the degrees of Z_i, D are at most T . A polynomial of degree at most $\delta(1+T)$ in q variables has at most $\binom{\delta(1+T)+q}{q}$ monomials. This number is polynomial for fixed q . Expanding and adding the explicit input monomials therefore takes polynomially many rational operations; coefficient bit lengths grow at most polynomially under these products and sums. Positivity of D^δ proves sign preservation. $\square$

For polyhedral limits we use Hoffman's bound [13]. Given fixed matrices M, T , there is a constant $H_{M,T}$ such that, whenever $K(a, b) = \{y : My \leq a, Ty = b\}$ is nonempty,

$$\text{dist}(y, K(a, b)) \leq H_{M,T}(\|(My - a)_+\| + \|Ty - b\|). \quad (11)$$

Any fixed choices of finite-dimensional norms can be used, with a changed constant. The constant is uniform in the right-hand sides. Equalities can be encoded as opposite inequalities. In the exact attainment proof, only existence of this constant for each instance is needed. The approximation proofs require quantitative encoding bounds and will establish them explicitly. Neither form of the argument grants a uniform constant when the constraint normals vary with the leader.

2 Exact global response compression

The positive quadratic curvature within each local block permits its elimination after fixing the aggregate and shared multipliers. The full follower can still have many stationary points and several global optima. We first encode all feasible stationary candidates, then compare their actual follower values. This comparison is what makes the method valid for a nonconvex aggregate cost.

The underlying geometry can be seen before introducing multipliers. Fix a leader x and an attainable aggregate w . On the fiber $\{z \in P(x) : U(x)z = w\}$ the aggregate cost is constant, and the remaining cost is a strictly convex quadratic. It has exactly one minimizer. A global follower optimum must be this minimizer on its own fiber. Thus aggregate coordinates determine a global response uniquely when that response exists. Constructing its formula still requires handling the many local rows and shared constraints; the multipliers below provide that construction without quantifying all follower variables.

Theorem 2.1 (Exact quadratic-block optimization). *For the model of section 1.2, fix (r, d, s, k_+, k_-) . There is an algorithm, polynomial in (L, δ) , that decides optimistic bilevel feasibility. If feasible, the upper minimum is attained and the algorithm returns an optimizer (x, z) and its value in one common real-algebraic field, with polynomial total encoding length. The number of blocks, their local constraint counts, and the number and support sizes of upper polynomials may grow. Consequently the algorithm is polynomial in L under the degree-encoding condition of section 1.4.*

The theorem includes scalar local quadratics as a special case. The aggregate polynomial need not be convex, and no linear independence or strict feasibility of the follower constraints is required.

Positive definiteness of each $Q_b(x)$ on C , compactness, and the stated coordinate bounds are hypotheses of the problem class. The following construction proves the theorem and also provides the response formula used in subsequent extensions.

2.1 Local quadratic responses

Write A_b, H_b for the block columns of the shared matrices. Introduce

$$\zeta = (x, w, \lambda, \mu) \in \mathbb{R}^{r+s+k_+=k_-}, \quad w = U(x)z, \quad (12)$$

where λ is unrestricted and $\mu \geq 0$. Define

$$q_b(\zeta) = c_b(x) + U_b(x)^\top \nabla_w \phi(x, w) + A_b^\top \lambda + H_b^\top \mu. \quad (13)$$

At a follower local minimizer, the negative gradient belongs to the normal cone of its polyhedron. To recall why no constraint qualification is needed, every feasible tangent direction has nonnegative objective derivative at a local minimum. The polar of the polyhedral tangent cone, by the linear alternative, is the span of the equality normals plus the cone generated by the active inequality normals. This gives multipliers for all rows, with nonnegative inequality multipliers and complementary slackness. Separating shared and local rows shows that each block satisfies the optimality conditions of

$$\min_{y \in P_b(x)} \frac{1}{2} y^\top Q_b(x) y + q_b(\zeta)^\top y. \quad (14)$$

For fixed ζ with $x \in C$, this is a strictly convex quadratic program on a bounded polyhedron. If feasible it has a unique minimizer. Its gradient agrees, at the proposed block response, with the original follower gradient after the shared multipliers have been added. This statement does not assert convexity of the original follower.

The scalar case makes the elimination transparent. If $Q_i(x) = a_i(x) > 0$ and the local set is $[\ell_i(x), u_i(x)]$, the response in (14) is

$$z_i = \text{clip}_{[\ell_i(x), u_i(x)]} \left(-\frac{q_i(\zeta)}{a_i(x)} \right). \quad (15)$$

The signs of $q_i + a_i \ell_i$ and $q_i + a_i u_i$ determine the branch. Choose the lower bound if the first expression is nonnegative, otherwise the upper bound if the second is nonpositive, and otherwise the interior formula $-q_i/a_i$. Boundary formulas agree, including when the interval is a singleton. Joint regimes are determined by realizable signs in (12), rather than by an independent choice among three states for every follower variable.

Lemma 2.2 (A polynomial list for each block). *For fixed d , one can construct polynomially many rational formulas for the solution of (14), each with a polynomial validity predicate in ζ . On its validity region in $x \in C$, each formula has a positive polynomial denominator. Every feasible local optimum is represented, and all valid formulas at the same ζ agree. Their degrees are $O(d\delta)$ and their coefficient bit lengths are polynomial in (L, δ) .*

Proof. Choose an independent row basis $\bar{E}_b$ of the fixed matrix E_b ; let its size be t_b . Retain all original equality rows for feasibility tests. Enumerate inequality subsets J for which $V_J = [\bar{E}_b; (G_b)_J]$ has independent rows. There are at most $\sum_{j=0}^{d_b-t_b} \binom{m_b}{j}$ such subsets, where m_b is the number of local inequalities. With $r_J(x)$ denoting the corresponding right-hand sides, solve

$$M_J(x) \begin{pmatrix} y \\ \theta \end{pmatrix} = \begin{pmatrix} -q_b(\zeta) \\ r_J(x) \end{pmatrix}, \quad M_J(x) = \begin{pmatrix} Q_b(x) & V_J^\top \\ V_J & 0 \end{pmatrix}. \quad (16)$$

This matrix is nonsingular on C . Indeed, a null vector obeys $Q_b y + V_J^\top \theta = 0$ and $V_J y = 0$, so $y^\top Q_b y = 0$. Positive definiteness gives $y = 0$, and row independence gives $\theta = 0$.

Let $\Delta_J = \det M_J$ and let P_J be the vector of Cramer numerators for (y, θ) . Use

$$D_J = \Delta_J^2, \quad Y_J = \Delta_J P_J, \quad (y, \theta) = Y_J / D_J.$$

Thus $D_J > 0$ on C . A branch is valid if its y satisfies every original local row and the entries of θ belonging to J are nonnegative. Multiplication by D_J turns these conditions into polynomial equalities and inequalities. Omitted inequality multipliers are zero. The conditions imply sufficient KKT conditions for the strictly convex local problem, proving soundness and agreement of valid branches.

For completeness, represent the negative gradient of the local minimizer by equality normals and a nonnegative combination of active inequality normals. Project onto the quotient by the equality row space. If the positive-support projected generators are linearly dependent, let a be a nonzero dependence, change its sign so some $a_i > 0$, and replace their positive coefficients θ_i by $\theta_i - ta_i$, where $t = \min_{a_i > 0} \theta_i / a_i$. All coefficients remain nonnegative, at least one vanishes, and the represented vector is unchanged in the quotient. Absorb the change in the equality multipliers and repeat. Eventually the surviving projected normals are independent. Together with $\bar{E}_b$ they form one of the enumerated V_J , and their stationarity system is (16).

The matrices have order at most $2d$. Their entries have degree $O(\delta)$ in a fixed number of variables, so determinants, adjugates, and cleared tests have degree $O(d\delta)$. Fixed matrix order and the explicit input encoding bound their expanded sizes and coefficient lengths polynomially. Rational linear algebra constructs $\bar{E}_b$ and tests independence in polynomial bit time. $\square$

2.2 One common description of the candidate graph

Collect all local branch-validity polynomials. Their total number is polynomial for fixed d , even if the number of rows in an individual block grows. Enumerate their realizable sign conditions in ζ , including zero signs, using the tools of section 1.5. A sign condition fixes which branches are valid in every block. Discard it if some block has no valid branch. For each remaining condition σ , choose the first valid branch of each block in a fixed ordering. The branches agree wherever they overlap by lemma 2.2, so this choice loses no response. A sign condition may have disconnected realization; its decisions still depend only on signs.

Let $D_{b\sigma}$ be the chosen denominator for block b , and write

$$D_\sigma = \prod_{b=1}^B D_{b\sigma}, \quad z_i = Z_{i\sigma}(\zeta) / D_\sigma(\zeta). \quad (17)$$

The polynomial numerator $Z_{i\sigma}$ is the chosen local numerator multiplied by the other block denominators. Empty products are one; empty blocks can be removed after checking their rows. In particular the construction also applies if there are no follower coordinates. Every D_σ is strictly positive on its declared region.

Define $K_\sigma(\zeta)$ to be its sign condition together with $x \in C$ and the following polynomial conditions:

$$\begin{aligned} U(x)Z_\sigma &= D_\sigma w, & AZ_\sigma &= D_\sigma b(x), \\ HZ_\sigma &\leq D_\sigma a(x), & \mu &\geq 0, \\ \mu_j((HZ_\sigma)_j - D_\sigma a_j(x)) &= 0 & (j &= 1, \dots, k_\leq). \end{aligned} \quad (18)$$

The local constraints and local multiplier signs already hold through the chosen valid branches. Thus K_σ describes exactly the follower-feasible KKT candidates encoded in that regime.

The numerator of the follower value is

$$T_\sigma = \sum_b \left(\frac{1}{2} Z_{b\sigma}^\top Q_b(x) Z_{b\sigma} + D_\sigma c_b(x)^\top Z_{b\sigma} \right) + D_\sigma^2 \phi(x, w). \quad (19)$$

Set $\xi = (w, \lambda, \mu, t)$ and define

$$\mathcal{E}(x, \xi) = \bigvee_\sigma \left[K_\sigma(x, w, \lambda, \mu) \wedge t D_\sigma^2 = T_\sigma \right]. \quad (20)$$

This is a quantifier-free formula in $r + s + k + 1$ variables. Its represented pairs consist only of feasible follower points with their actual values; every global follower minimizer has such an encoding by polyhedral first-order necessity. There may be several multiplier encodings of the same response and several nonglobal candidates.

Lemma 2.3 (Polynomial construction size). *The regime list, the polynomial response functions (17), and (20) can be constructed in polynomial bit time in (L, δ) for the fixed structural dimensions. Substituting these response functions in all upper data preserves this bound.*

Proof. The number of local subsets and tests is polynomial for fixed d . The ambient dimension of their sign conditions is $r + s + k$, fixed, and their degrees are $O(d\delta)$. Hence the realizable sign list and its explicit conjunctions have polynomial size and construction cost. No Cartesian product of local lists is taken.

Products of the selected denominators have degree $O(Bd\delta)$. The numerators $Z_{i\sigma}$ have the same order of degree, with an additional $O(d\delta)$ term when $B = 0$ is included in the bound. Equations (18) and (19) have degree $O((B + 1)d\delta)$. Each such polynomial has a fixed number of variables, so its dense monomial count is polynomial in that degree. Repeated multiplication adds coefficient bit lengths and logarithmic convolution counts, giving polynomial coefficient lengths as well. For example, after clearing rational coefficient denominators, a product of B factors with at most M monomials and coefficient magnitudes at most 2^H has coefficients bounded by $M^B 2^{BH}$, whose bit lengths are $O(B(H + \log M))$. Rational denominator clearing has the same polynomial bound.

Finally lemma 1.1 applies separately to every explicitly listed upper polynomial, giving degree $O((B + 1)d\delta^2)$ and polynomial total size. Positive denominators preserve all weak and strict inequality signs. The objective equation for a new value coordinate η is treated as $\eta D_\sigma^\delta - D_\sigma^\delta F(x, Z_\sigma/D_\sigma) = 0$. The added coordinate is a fixed increase in dimension. $\square$

2.3 Globality, optimization, and algebraic recovery

The exact global-response predicate is

$$\mathcal{R}(x, \xi) = \mathcal{E}(x, \xi) \wedge \forall \xi' [\mathcal{E}(x, \xi') \implies t \leq t']. \quad (21)$$

There are just two fixed-dimensional candidate copies. If $P(x) \neq \emptyset$, a global minimum exists and has an encoding in $\mathcal{E}$. Therefore any candidate passing the comparison has value at most this global minimum, and, since it is itself feasible, has equal value. Conversely, a global optimum cannot exceed the value of any feasible candidate. Thus (21) represents *every and only* global responses, including all ties.

An explicit upper predicate $\mathcal{U}(x, \xi, \eta)$ is obtained by a single disjunction over regimes: in regime σ impose the substituted upper rows and $\eta = F(x, Z_\sigma/D_\sigma)$, clearing positive denominators. Use

the regime conditions of K_σ , so its decoded vector is the one encoded by $\mathcal{E}$. The optimistic feasible value relation is

$$\Psi(x, \xi, \eta) = \mathcal{R}(x, \xi) \wedge \mathcal{U}(x, \xi, \eta). \quad (22)$$

There is one copy of the quantified global comparison, outside the upper regime disjunction. One may eliminate its quantifier once to obtain a polynomial-size quantifier-free $\mathcal{R}_0$ before any further construction. In particular, a growing number of regimes or upper rows does not introduce a growing number of real variables.

Quantifier elimination decides whether (22) has a solution. To recover an attained optimum jointly, use

$$\Psi(x, \xi, \eta) \wedge \neg \exists \bar{x}, \bar{\xi}, \bar{\eta} [\Psi(\bar{x}, \bar{\xi}, \bar{\eta}) \wedge \bar{\eta} < \eta]. \quad (23)$$

After replacing $\mathcal{R}$ by $\mathcal{R}_0$, this still uses only a fixed number of variables and has polynomial length, degrees, and coefficient bit lengths. Algebraic sampling after elimination provides (x, ξ, η) in one field of polynomial degree and encoding length. Testing its regime signs selects a valid rational decoding formula; evaluating every $Z_{i\sigma}/D_\sigma$ in that same field recovers z . Polynomial degree and coefficient bounds ensure polynomial bit cost for these evaluations and the total output. No product of N unrelated algebraic extensions is formed.

Lemma 2.4 (Closed response graph with fixed normals). *For the fixed-normal model of section 1.2, the graph $\{(x, z) : x \in C, z \in S(x)\}$ is compact. Consequently $\mathcal{O}$ is compact and an optimistic optimum is attained whenever $\mathcal{O} \neq \emptyset$.*

Proof. Continuity of the coordinate bounds on compact C gives uniform boundedness. Consider a convergent sequence (x_n, z_n) in the graph, with limit (x, z) . Its limit is follower-feasible by continuity of all rows, so $P(x)$ is nonempty. Fix $y \in P(x)$. All local, shared, and box normals form one matrix independent of the leader. The systems $P(x_n)$ are nonempty because they contain z_n . Hoffman's bound (11), applied to y in each such system, gives $y_n \in P(x_n)$ with $y_n \rightarrow y$: the residual at y tends to zero as the polynomial right-hand sides converge. Since $f(x_n, z_n) \leq f(x_n, y_n)$, continuity implies $f(x, z) \leq f(x, y)$. This holds for every feasible y , so $z \in S(x)$. The graph is closed and bounded over compact C , hence compact. Upper weak polynomial rows define a closed subset, and F is continuous. $\square$

Lemma 2.4 makes (23) nonempty whenever the optimistic problem is feasible. Together with lemmas 2.2 and 2.3 and the exact global comparison, this completes the proof of theorem 2.1. The response formula uses a fixed number of shared quantities rather than a fixed follower dimension. The local multiplier-cell principle is classical [18]; global comparison and upper optimization are the additional steps needed here.

2.4 Moving normals and changing local ranks

Now allow $E_b(x), G_b(x), A(x), H(x)$ to be polynomial matrices, preserving the same row-count restrictions, positive-definite local quadratics, compact C , and the polynomial coordinate bounds. Every follower feasible set is still a compact polyhedron at a fixed leader, but its geometry can change discontinuously with x .

Lemma 2.5 (Rank changes preserve compression). *The moving-normal model has a polynomial-size exact global-response formula with the same fixed total dimension and common-field decoding as (21).*

Proof. Replace A_b, H_b by $A_b(x), H_b(x)$ in (13). For a local block enumerate all pairs (I, J) of equality and inequality row subsets with $|I| + |J| \leq d_b$, including the empty pair. Put $V_{I,J}(x) = [(E_b(x))_I; (G_b(x))_J]$ and form the KKT matrix as in (16). Use the branch only where $\Delta_{I,J}(x)^2 > 0$, and again multiply the Cramer numerators by $\Delta_{I,J}$ to obtain this positive denominator. Validity also tests all original local rows and the nonnegative multipliers for J .

A valid branch satisfies local KKT conditions with zero multipliers on omitted rows, so it is the unique local minimizer. Soundness does not require that I span all equality rows. For completeness, at any fixed leader select I to be a basis of its equality row space, whose rank may depend on that leader. Reduce the conic support of active inequality normals modulo this row space exactly as in lemma 2.2. This supplies an enumerated J for which $V_{I,J}(x)$ is independent. Positive definiteness makes its KKT matrix nonsingular, and its branch returns the optimum.

There are polynomially many pairs for fixed d . Their determinant nonvanishing tests are included among the sign polynomials, so all clearing operations occur only where the chosen denominators are positive. The sign-list selection, product denominators, resource consistency and global comparison proceed as before. Row counts and rank changes do not add compressed variables. Pointwise polyhedral KKT necessity still covers every global minimum. The degree and encoding bounds in lemma 2.3 remain valid, since multiplying by moving polynomial normals changes them by only $O(\delta)$. $\square$

Example 2.6 (Optimistic nonattainment with a moving row). Let $x, z \in [0, 1]$, require $xz = 0$, and let the follower minimize $(z - 1)^2$. The response is $z = 0$ for $x > 0$ and $z = 1$ for $x = 0$. For upper objective $F = x + z$, the infimum is zero and is not attained. This is a moving shared equality, a one-dimensional positive local quadratic, and compact uniform boxes. Interpreting the same equality as a local row also illustrates lemma 2.5: for $x \neq 0$ its equality-active KKT determinant is nonzero; at $x = 0$ that determinant vanishes and the empty equality basis supplies the response $z = 1$. Dropping the determinant guard or fixing the equality basis would respectively introduce an invalid branch or lose a response.

2.5 Pessimistic values and attainment decisions

Theorem 2.7 (Exact infima and pessimistic optimization). *For either the fixed-normal or moving-normal quadratic-block model, with the same fixed structural dimensions and encoding, optimistic feasibility, its exact infimum, and attainment are decidable in polynomial bit time in (L, δ) . The same holds for pessimistic feasibility and value under (8)–(9). If the infimum is attained, the algorithm returns an optimal leader and a corresponding response, worst for the upper objective in the pessimistic case, in one common field of polynomial total encoding size. Every feasible problem has finite value. Fixed-normal optimistic attainment is guaranteed by lemma 2.4; no unconditional pessimistic attainment assertion is made.*

Proof. Use a quantifier-free exact response predicate $\mathcal{R}_0(x, \xi)$ from (21) or lemma 2.5. Let $\mathcal{F}(x, \xi, \eta)$ denote the regime-wise, denominator-cleared equation for the upper value, and let $\mathcal{B}(x, \xi)$ denote the disjunction of violated upper rows in their respective regimes. Both use a single regime disjunction and have polynomial size. Define

$$\begin{aligned} T(x, \eta) &\iff \exists \xi [\mathcal{R}_0(x, \xi) \wedge \mathcal{F}(x, \xi, \eta)], \\ \text{Bad}(x) &\iff \exists \xi [\mathcal{R}_0(x, \xi) \wedge \mathcal{B}(x, \xi)], \\ D(x) &\iff x \in C \wedge (\exists \xi \mathcal{R}_0(x, \xi)) \wedge \neg \text{Bad}(x). \end{aligned} \tag{24}$$

These are exact: the existence conjunct excludes empty follower sets, and Bad detects any globally optimal response violating any upper row. It never filters the follower set by those rows. At a fixed feasible leader $S(x)$ is compact, so the maximum of F exists. Its graph on the pessimistic domain is

$$\text{Worst}(x, \eta) \iff D(x) \wedge T(x, \eta) \wedge \forall \eta' [T(x, \eta') \implies \eta' \leq \eta]. \quad (25)$$

One may eliminate T, D before using this expression, or expand a fixed number of copies with fresh internal variables. Thus the number of variables stays fixed even if there are many upper constraints.

Elimination describes the attainable worst values $V = \{\eta : \exists x \text{ Worst}(x, \eta)\}$ as a univariate semialgebraic set. It also decides whether D is empty. Uniform boundedness of z and compactness of C bound F on all feasible pairs, even if the response graph is not closed. Hence nonempty V has a finite infimum. If $V_0(\eta)$ is its quantifier-free description, that infimum η_0 is characterized exactly by

$$\begin{aligned} I(\eta_0) \iff & \forall \eta [V_0(\eta) \implies \eta_0 \leq \eta] \quad \wedge \\ & \forall \varepsilon [\varepsilon > 0 \implies \exists \eta (V_0(\eta) \wedge \eta < \eta_0 + \varepsilon)]. \end{aligned} \quad (26)$$

It has a unique solution when V is nonempty and uses a fixed number of variables. Univariate algebra or elimination and sampling computes η_0 exactly. Membership $V_0(\eta_0)$ decides attainment. When attained, sample the joint formula

$$I(\eta_0) \wedge \text{Worst}(x, \eta_0) \wedge \mathcal{R}_0(x, \xi) \wedge \mathcal{F}(x, \xi, \eta_0).$$

This returns the leader, its value, and a worst global response in one field, after the rational regime decoding. Multiple multiplier encodings have no effect on the represented upper values or feasibility. For the optimistic variant replace V by the attainable upper values of (22), and use its relation in the joint sample. All formulas satisfy the polynomial bounds already proved. If the endpoint is not in V , it is reported as an unattained infimum, not as an optimizer. $\square$

Example 2.8 (Fixed normals do not ensure pessimistic attainment). Let $x, z \in [0, 1]$ and

$$f(x, z) = z^2(1 - z)^2 + xz, \quad F(x, z) = x + z.$$

This is the model (3) with local term $z^2/2$, $w = z$, and aggregate $\phi(x, w) = w^2(1 - w)^2 - w^2/2 + xw$. For $x > 0$ the unique optimum is $z = 0$; for $x = 0$ the global optima are exactly $z = 0, 1$. Thus $W_0(x) = x$ for $x > 0$, while $W_0(0) = 1$. The pessimistic infimum is zero and unattained. Adding the upper row $z \leq 1/2$ makes precisely the leaders $x > 0$ pessimistically admissible, whereas the optimistic model can use $(x, z) = (0, 0)$. At $x = 0$ the stationary point $z = 1/2$ has cost $1/16$ and is rejected by (21); the two zero-cost endpoints survive. This example separates nonglobal stationarity, true ties, and universal upper feasibility within the same bounded instance.

Corollary 2.9 (Bounded algebraic degree with constant local Hessians). *In the fixed-normal model, suppose additionally that every Q_b is independent of x . If the structural dimensions and input degree δ are fixed, the algebraic degree of a common-field optimizer and value in theorem 2.1 is bounded by a constant independent of the number of follower variables, local or upper rows, and input coefficient bit lengths. The same degree bound holds for the infimum in theorem 2.7, and jointly for an optimal leader and worst response when the pessimistic infimum is attained. Total output encoding length remains polynomial in L .*

Proof. Each selected KKT matrix in (16) is a constant rational matrix. Its inverse has constant rational entries, so every local response is a polynomial in ζ of degree $O(\delta)$; only its coefficient lengths

and the number of regimes depend on the number of blocks or rows. Summing block contributions does not raise degree. Thus the candidate equations and substituted upper polynomials have degree $O(\delta^2)$, independent of B and N . There is no need to use a growing-degree product denominator: all denominators are rational constants.

The degree bounds for quantifier elimination in [3, Theorem 1.3.1] depend on the input degrees and variable-block dimensions, not on the number of polynomials. Its simultaneous sample-point construction likewise bounds the degree of each univariate representation in terms of polynomial degree and ambient dimension [3, Section 3.1.3]. Apply these bounds to the fixed-number-of-variables formulas (23), (25), and (26). The required number of elimination and sampling steps is fixed, so the resulting degree depends only on δ and the structural dimensions. Polynomial evaluation of the responses stays in the sampled field. The bit bounds already proved control the growing coefficient and output lengths. $\square$

2.6 A rational LP specialization for supplied low-rank coupling

When both the response formulas and the upper data are affine, linear programming replaces quantifier elimination.

Corollary 2.10 (Supplied diagonal-plus-fixed-rank Hessian). *Fix r and s . Let $X \subset [0, 1]^r$ be a rational polytope and suppose the constant rational follower Hessian is supplied in the form*

$$Q = \text{diag}(d_1, \dots, d_N) + VMV^\top \succ 0, \quad d_i > 0, \quad V \in \mathbb{Q}^{N \times s}, \quad M = M^\top \in \mathbb{Q}^{s \times s}.$$

For leader $x \in X$ the follower minimizes $\frac{1}{2}z^\top Qz + (c + Cx)^\top z$ on $[0, 1]^N$. With any explicit rational affine upper rows and objective, feasibility and an attained global optimum, when feasible, are computable in polynomial bit time, with a rational optimizer and value. No bound on the condition number or magnitude of M is needed, and M need not be positive semidefinite. The decomposition is part of the input; this is not an algorithm for finding one.

Proof. Let $w = V^\top z$. The unique follower optimum satisfies

$$z_i = \text{clip}_{[0,1]} \left(-\frac{c_i + C_i x + V_i M w}{d_i} \right).$$

The two clipping thresholds per coordinate are affine hyperplanes in $(x, w) \in \mathbb{R}^{r+s}$. Enumerate their realizable cells, including all faces and constant threshold cases. On each cell a clipped response is a specified affine function; it extends with the same values to the cell closure because adjacent formulas agree at a threshold. The closure is given by the corresponding weak inequalities and the cell equalities. Impose $w = V^\top z$ after affine substitution, along with the upper rows, $x \in X$, and bounds

$$\sum_i \min(V_{i\ell}, 0) \leq w_\ell \leq \sum_i \max(V_{i\ell}, 0) \quad (\ell = 1, \dots, s).$$

This is a bounded rational LP in fixed dimension. Every feasible point satisfies the complete box KKT conditions for the SPD follower and thus is its unique global response. Conversely every leader–response pair is included in a cell LP. There are polynomially many cells in fixed $r + s$, and rational LP optimization and response reconstruction have polynomial bit cost. Taking the best of the finitely many attained cell optima proves the claim. Uniqueness makes optimistic and pessimistic selection identical in this specialization. $\square$

Positive local quadratics are what keep all decoded follower coordinates in one field generated by fixed-dimensional data. General polynomial local costs need a different output and accuracy analysis. Even independent cubic costs $z_i^3/3 - a_i z_i$ on $[1, a_i + 1]$, for integers $a_i \geq 1$, have unique responses $\sqrt{a_i}$. An upper inequality on their sum therefore embeds exact square-root-sum comparison. This observation motivates the approximation results; it does not label that arithmetic problem NP-hard. Detailed output and structural boundaries are treated separately. A related support-function elimination theorem for polyhedral blocks is proved in appendix A; it clarifies why many local variables can be recovered without giving each one a quantified coordinate.

3 Robust responses and certified dense recovery

The response description admits two different extensions. First, the leader can protect itself against followers whose cost is close to the true optimum, without requiring those followers to be stationary. Second, a structured surrogate can identify enough coordinate statuses to recover the exact response of a dense convex quadratic follower. The first extension changes the set of responses considered by the leader; the second preserves the original follower and certifies its solution.

3.1 Robustness through criterion-specific measurements

Use the moving-normal quadratic-block model of section 2.4, including its compact domain C , polynomial coordinate bounds, and positive-definite local quadratic matrices. The fixed-normal model is included. The follower value $v(x)$ and cost-near-optimal set $S_\rho(x)$ are those of (5) and (10), with a supplied polynomial budget $\rho(x) \geq 0$ on C . Let each upper criterion, including the objective indexed by zero, have the form

$$G_j(x, z) = p_j(x, T_j(x)z), \quad j = 0, \dots, m, \quad (27)$$

where $T_j(x)$ has $h_j \leq h$ rows and p_j, T_j are explicitly encoded rational polynomials. Only h is fixed; the number of criteria and the rank of their combined measurement matrices may grow. An affine upper row needs only one measurement, regardless of its follower support. Leader-only terms belong directly to p_j . Affine measurements with an additional offset can be written in this form by absorbing the offset into p_j .

The robust feasible leaders and objective are

$$D_\rho = \{x \in C : P(x) \neq \emptyset, G_j(x, z) \leq 0 \text{ for every } z \in S_\rho(x), j = 1, \dots, m\}, \quad (28)$$

$$W_\rho(x) = \max_{z \in S_\rho(x)} G_0(x, z), \quad \text{val}_\rho = \inf_{x \in D_\rho} W_\rho(x). \quad (29)$$

The budget is measured from the global follower minimum. Upper rows never remove responses from $S_\rho(x)$ before an adversarial choice. The near-optimality model and separate upper-row adversaries are established in [6]; related complexity-class membership results and an objective-robust variant appear in [5]. The following theorem concerns exact global optimization for the stated structural dimensions.

Theorem 3.1 (Exact near-optimal robustness). *Fix (r, d, s, k_-, k_+, h) . For (28)–(29), feasibility, the finite infimum when feasible, and attainment are computable in polynomial bit time in (L, δ) . If attained, an optimal leader and a worst near-optimal response can be returned jointly in one real-algebraic field with polynomial total encoding length. At any supplied rational leader with $P(x) \neq \emptyset$, the exact worst value and a witnessing near-optimal response for any specified criterion can also be computed in polynomial bit time. For many separately requested criteria, the polynomial encoding*

guarantee applies separately to each witness; a common field containing all unrelated witnesses is not promised.

Lemma 3.2 (Measurement-fiber threshold). *Fix one measurement matrix $T(x)$ with at most h rows. Let*

$$P_T(x, \tau) = \{z \in P(x) : T(x)z = \tau\}, \quad m_T(x, \tau) = \min_{z \in P_T(x, \tau)} f(x, z)$$

when the fiber is nonempty. At a follower-feasible leader,

$$\tau \in T(x)S_\rho(x) \iff P_T(x, \tau) \neq \emptyset \text{ and } m_T(x, \tau) \leq v(x) + \rho(x). \quad (30)$$

Moreover the right side is equivalent to existence of a feasible KKT candidate of the measurement-fiber problem whose cost is at most $v(x) + \rho(x)$. That candidate need not be a stationary point of the unrestricted follower problem.

Proof. If a near-optimal response has measurement τ , minimizing on its compact fiber cannot increase cost. Conversely, a fiber minimizer whose cost meets the budget is itself a near-optimal response with that measurement. A global fiber minimizer satisfies polyhedral KKT conditions, so it supplies the asserted candidate whenever the right side holds. Conversely every feasible fiber candidate meeting the budget is already a near-optimal point. Its global optimality on the fiber is unnecessary for this converse. Global comparison remains essential in computing the nominal value $v(x)$. $\square$

Proof of theorem 3.1. If C is empty, report infeasibility. Otherwise first obtain the nominal value graph $\mathcal{V}(x, v)$ by eliminating the response variables from the global predicate (21). It is quantifier-free of polynomial size, and it has exactly one value $v = v(x)$ at a feasible leader and none otherwise.

For one criterion j , add $T_j(x)z = \tau$ as h_j shared equalities. The parameters are now (x, τ) , of fixed dimension $r + h_j$. All possible measurements lie in a rational box with polynomial encoding length: maximize the absolute values of the polynomial entries of T_j and the coordinate bounds on compact C , using fixed-dimensional real algebra, and round the resulting bounds outwards to integers. Summing the products of these bounds gives a bound for each measurement. Univariate root bounds applied to the polynomial-size algebraic outputs ensure the integer bounds have polynomial bit length. Thus the enlarged parameter domain can be taken compact. The positive-definite local quadratics are unchanged, and the added normals are polynomial, so lemma 2.5 applies.

Let $\mathcal{E}_j^{\text{fib}}(x, \tau, \xi_j)$ be the quantifier-free candidate predicate constructed as in (20), including the new equality multipliers. Its coordinate t_j is the candidate's actual follower cost. Define

$$\mathcal{A}_j(x, v, q) \iff \exists \tau, \xi_j [\mathcal{E}_j^{\text{fib}}(x, \tau, \xi_j) \wedge t_j \leq v + \rho(x) \wedge q = p_j(x, \tau)]. \quad (31)$$

By lemma 3.2, at $v = v(x)$ this describes exactly all attainable criterion values on $S_\rho(x)$. A representative may be a minimizer on its measurement fiber even when the response it replaces is nonstationary for the unrestricted follower.

Eliminate the variables in (31) *separately for every criterion*, producing quantifier-free $\mathcal{A}_j$. Then eliminate q in

$$\mathcal{B}_j(x, v) \iff \exists q [\mathcal{A}_j(x, v, q) \wedge q > 0].$$

Each elimination has a fixed total number of variables. There are polynomially many such computations, whose output formulas have polynomial total length and degree. After this elimination, the formula

$$\mathcal{D}(x) \iff x \in C \wedge \exists v [\mathcal{V}(x, v) \wedge \bigwedge_{j=1}^m \neg \mathcal{B}_j(x, v)] \quad (32)$$

uses only one shared nominal-value variable. The empty-follower case is excluded by $\mathcal{V}$. Introducing simultaneous quantified adversaries for all rows would not give the claimed bound.

Since $S_\rho(x)$ is nonempty and compact at every feasible leader, its continuous criterion attains a maximum. The worst-objective graph is

$$\mathcal{W}(x, q) \iff \mathcal{D}(x) \wedge \exists v [\mathcal{V}(x, v) \wedge \mathcal{A}_0(x, v, q) \wedge \neg \exists q' (\mathcal{A}_0(x, v, q') \wedge q' > q)]. \quad (33)$$

Eliminate $\mathcal{D}$ first if desired; every formula still has a fixed number of variables. The attained worst-value set is bounded by uniform follower bounds and compact C . Its emptiness, exact infimum, and endpoint membership follow by the construction (26). Endpoint membership decides attainment.

When attained, sample the leader, nominal value and worst value together with *one* measurement and a thresholded fiber candidate realizing that worst value, using the uneliminated relation (31) for $j = 0$. This adds only a fixed number of variables. Simultaneous sampling returns a common field of polynomial degree and encoding length. The rational block formulas of the fiber candidate recover all follower coordinates in that field. Their cost meets the true budget and their criterion equals the exact worst value. The same procedure at a supplied rational follower-feasible leader computes one criterion maximum and witness, including a violating witness whenever that maximum is positive. Repeating it for different criteria gives separate encodings and does not require a joint field for all those witnesses. $\square$

A bounded budget can be an additional leader coordinate. For example, choosing $G_0(x, z) = -\rho$ maximizes tolerated follower cost loss under all robust upper rows. This adds one fixed leader dimension. Exact rational adversarial output is generally impossible: minimizing $z^2/2$ on $[0, 1]$ with budget $1/4$ gives near-optimal interval $[0, 1/\sqrt{2}]$, whose worst response for criterion z is irrational.

Corollary 3.3 (Constant local matrices and algebraic degree). *If the local Q_b, E_b, G_b are independent of the leader, the common-field outputs of theorem 3.1 have degree bounded solely by input degree and the fixed structural dimensions. This includes polynomially moving shared normals and measurement matrices $T_j(x)$. For fixed input degree, the bound is independent of the follower and criterion counts and coefficient bit lengths. It applies to one jointly recovered leader and adverse witness, or to each separately requested criterion witness. It does not assert attainment.*

Proof. Shared normals and added measurement rows enter the effective linear cost and shared consistency equations; they do not enter the local KKT matrix (16). That matrix is constant under the stated assumptions, so each local response is polynomial in the compressed variables of degree $O(\delta)$ with rational coefficients. All candidate, threshold and criterion formulas have degree bounded in terms of δ , independent of the numbers of rows and blocks. The degree-independent-of-polynomial-count elimination and sampling bounds used in corollary 2.9 apply to each separate criterion computation and then to (33) and one witness. The number of successive elimination and sampling steps along each dependency chain is fixed. Rational decoding preserves the resulting field. $\square$

3.2 When the robust optimum is attained

Proposition 3.4 (Convex fixed-normal attainment). *Under the hypotheses of theorem 3.1, suppose all follower constraint normals are fixed and $\phi(x, \cdot)$ is convex for each $x \in C$. Then every feasible robust problem attains its minimum, including when the nonnegative budget vanishes at some leaders.*

Proof. Let $C_F = \{x \in C : P(x) \neq \emptyset\}$. It is compact: any sequence of feasible leaders has, along a convergent subsequence, bounded feasible followers whose limit remains feasible. Fixed normals and (11) let every $z \in P(x)$ be approximated by feasible $z_n \in P(x_n)$ whenever $x_n \rightarrow x$ in C_F . Closed constraints and uniform bounds give the converse limit property.

The full follower objective is strictly convex in z , so it has one nominal minimizer $z^*(x)$. The comparison-and-repair argument in lemma 2.4, together with uniqueness, implies continuity of z^* on C_F , and hence of $v(x) = f(x, z^*(x))$. The graph of S_ρ is closed by continuity of f, v, ρ and the constraints, and its values are uniformly bounded.

To prove the needed approximation property for S_ρ , fix $x_n \rightarrow x$ in C_F and $z \in S_\rho(x)$. If $\rho(x) = 0$, then $z = z^*(x)$ and the nearby nominal minimizers supply near-optimal approximations. If $\rho(x) > 0$, for any $\theta \in (0, 1)$ the point $z_\theta = (1 - \theta)z + \theta z^*(x)$ is feasible and convexity gives

$$f(x, z_\theta) \leq (1 - \theta)f(x, z) + \theta v(x) \leq v(x) + (1 - \theta)\rho(x) < v(x) + \rho(x).$$

Repair z_θ into $P(x_n)$ by Hoffman. Continuity of the cost, nominal value and budget makes the repaired point near-optimal for all sufficiently large n . Taking θ successively to zero and using a diagonal sequence gives $z_n \in S_\rho(x_n)$ with $z_n \rightarrow z$.

For any continuous criterion, the upper limit of its near-optimal maximum is bounded by its maximum at x : choose maximizing responses and use a convergent subsequence and the closed graph. The reverse inequality follows by approximating a maximizing response at x using the preceding construction. Thus each criterion maximum is continuous on C_F . The robust upper rows define a closed subset of compact C_F , and the continuous worst objective attains its minimum there. $\square$

Example 3.5 (A positive budget does not ensure attainment). Set $\rho = 1/16$, $h(z) = 3z^2 - 2z^3$, and, for $c \in [\rho, 1/8]$, let

$$f_c(z) = z^2(1 - z)^2 + ch(z), \quad z \in [0, 1].$$

Both summands are nonnegative and $h(z) > 0$ for $z > 0$, so the unique nominal optimum is zero with value zero. Separating off $z^2/2$ puts this model in the positive-local-quadratic class with a scalar polynomial aggregate and fixed box normals. Direct factorization gives

$$f_\rho(z) - \rho = (z - 1)^2(z^2 - z/8 - 1/16).$$

Consequently $S_\rho(\rho) = [0, a] \cup \{1\}$, where $a = (1 + \sqrt{17})/16 < 1/3$. For $c > \rho$, the near-optimal set is $[0, a_c]$, with $0 < a_c < a$: the derivative $f'_c(z) = 2z(1 - z)(1 - 2z + 3c)$ changes sign once in $(0, 1)$, and $f_c(1) = c > \rho$. Its first crossing of the budget is therefore its only sublevel endpoint, and $a_c \rightarrow a$ as $c \downarrow \rho$.

If the leader minimizes c subject to $z \leq 1/2$ for every near-optimal response, its feasible set is $(\rho, 1/8]$ and its infimum ρ is unattained. Even without upper rows, put $c = \rho + x$ for $x \in [0, \rho]$ and minimize the worst value of $4x + z$. Its value is one at $x = 0$ and $4x + a_{\rho+x}$ for $x > 0$. On $0 < z \leq a < 1/3$,

$$f'_\rho(z) = z(4z^2 - 51z/8 + 19/8) > z/4, \quad h(z) < z.$$

Writing $c = \rho + x$, the identity $xh(a_c) = f_\rho(a) - f_\rho(a_c)$ implies $xa_c > (a - a_c)a_c/4$, and thus $4x + a_c > a$. The objective tends to a as $x \downarrow 0$, so this unconstrained robust infimum is also unattained. Unique nominal optima and a strictly positive budget therefore do not replace the convexity assumption in proposition 3.4.

Proposition 3.6 (An unrestricted quadratic criterion is hard). *Without a bound on the measurement dimension, exact worst-criterion computation is NP-hard even for a leader-free diagonal quadratic follower with unit-box constraints and no aggregate or resource coupling. Universal upper-threshold feasibility is coNP-complete on the following restricted family.*

Proof. For a graph on N vertices, let $f(z) = \sum_i z_i^2/2$ on $[0, 1]^N$ and take budget $\rho = N/2$. The nominal value is zero, so every point of the cube is near-optimal. Set

$$G(z) = \sum_{ij \in E} (z_i - z_j)^2.$$

This is convex. Every cube point is a convex combination of binary vertices, so its criterion is no greater than the maximum criterion of a binary vertex. At such a vertex, G is the cut size. Unweighted Max-Cut threshold decision is NP-complete [10]. Thus computing the worst value is NP-hard, and the robust row $G(z) \leq K - 1$ is satisfied for all responses exactly when there is no cut of size at least K . A violating binary vertex is a polynomial certificate for the complement on this family, proving coNP-completeness. This is an elementary transfer of the classical cut problem. A growing collection of separately checked affine rows is not an arbitrary quadratic criterion in a growing number of joint measurements. $\square$

3.3 A structured surrogate for a dense quadratic follower

We now consider a different model. Let $X \subset \mathbb{R}^r$ be a nonempty compact rational polytope. The true follower response is

$$z(x) = \arg \min_{0 \leq z \leq 1} \left\{ \frac{1}{2} z^\top Q z + (c + Cx)^\top z \right\}, \quad Q = Q^\top \succ 0. \quad (34)$$

All data are rational and the upper objective and rows are affine in $(x, z(x))$. Equalities can be represented by two rows. Supply a surrogate

$$\hat{Q} = \text{diag}(d) + VMV^\top \succ 0, \quad d_i > 0, \quad V \in \mathbb{Q}^{N \times s_0}, \quad M = M^\top \in \mathbb{Q}^{s_0 \times s_0},$$

with the same box and linear cost. Write $y(x)$ for its exact response and $E = Q - \hat{Q}$. No norm or rank bound on E is needed for correctness. The fixed-rank construction of corollary 2.10 provides a finite closed rational polyhedral cover of the surrogate response graph in $v = (x, w)$, $w = V^\top y(x)$. Each nonempty cover polytope P is compact, projects into X , and has an affine response formula $y_P(v)$. All of its points are consistent surrogate responses, and every leader has a graph point in the cover. The method accepts any supplied cover with these properties, including subdivisions. Its size and description length are part of the complexity bound.

Surrogate-based safe elimination has a direct precedent in the structured dictionary screening of Dantas and Gribonval [8]. Variational-inequality enclosures precede it in [16], and complete parameter-region analysis of QP algorithms appears in [2]. Here the target is an exact global upper optimum for the true dense follower, with the exponential enumeration confined to statuses not certified on a whole surrogate cell.

Lemma 3.7 (Directional response enclosure). *At a fixed leader put $a = c + Cx$, $e = z - y$, $p = Ey$, and $\hat{g} = \hat{Q}y + a$. Then*

$$e^\top Q e + p^\top e \leq 0, \quad (35)$$

and for any signed vector $b \in \mathbb{R}^N$,

$$\left| b^\top e + \frac{1}{2} b^\top Q^{-1} p \right| \leq \frac{1}{2} \sqrt{(b^\top Q^{-1} b)(p^\top Q^{-1} p)}. \quad (36)$$

Proof. The two box variational inequalities are $(Qz + a)^\top(y - z) \geq 0$ and $(\hat{Q}y + a)^\top(z - y) \geq 0$. Combining the two inequalities gives (35). Completing the square yields

$$(e + \tfrac{1}{2}Q^{-1}p)^\top Q(e + \tfrac{1}{2}Q^{-1}p) \leq \tfrac{1}{4}p^\top Q^{-1}p.$$

Cauchy–Schwarz in the Q norm gives (36). This is an enclosure: points inside the ellipsoid need not be realizable follower responses. It uses the exact surrogate response, not an unchecked numerical approximation. $\square$

3.4 Certificates on a whole response cell

On a nonempty cover polytope P , the functions $y(v)$ and $p(v) = Ey(v)$ are affine. Compute the rational number

$$\eta_P = \max_{v \in \text{vert}(P)} p(v)^\top Q^{-1}p(v). \quad (37)$$

It is also the maximum over P , since the quadratic is convex and P is the convex hull of its vertices. In fixed ambient dimension $r + s_0$, all vertices can be enumerated by rational row-basis tests, including lower-dimensional cells and singletons, as in appendix A.1. Define

$$\begin{aligned} s_i(v) &= y_i(v) - \tfrac{1}{2}(Q^{-1}p(v))_i, & R_i &= \tfrac{1}{2}\sqrt{(Q^{-1})_{ii}\eta_P}, \\ t_i(v) &= \hat{g}_i(v) + \tfrac{1}{2}p_i(v), & S_i &= \tfrac{1}{2}\sqrt{Q_{ii}\eta_P}. \end{aligned} \quad (38)$$

Applying (36) to a coordinate vector and to its image under Q gives, throughout P ,

$$s_i - R_i \leq z_i(x) \leq s_i + R_i, \quad t_i - S_i \leq (Qz(x) + a(x))_i \leq t_i + S_i. \quad (39)$$

We use labels L, U, F to certify, respectively, $z_i = 0$, $z_i = 1$, and zero true gradient in coordinate i throughout the cell. The label F is a stationarity equation; it allows a zero-gradient bound point. The following tests are sufficient:

Label	Whole-cell test
L	$\min_P t_i > S_i$ or $\max_P s_i + R_i \leq 0$
U	$\max_P t_i < -S_i$ or $\min_P s_i - R_i \geq 1$
F	$\min_P s_i > R_i$ and $\max_P s_i < 1 - R_i$

A positive gradient in a box optimum forces the lower bound, a negative gradient forces the upper bound, and strict interiority forces zero gradient. Zero gradient alone does not force either bound; hence the strict signs in the first tests for L and U . All affine extrema are attained at the already enumerated vertices. Square-root comparisons are exact rational tests: for rational A and $B \geq 0$, for instance, $A > \sqrt{B}/2$ means $A > 0$ and $4A^2 > B$.

The following boundary refinement avoids unnecessary ambiguity on closed cells. The weaker conditions

$$\min_P s_i \geq R_i, \quad \max_P s_i \leq 1 - R_i, \quad \max_P s_i > R_i, \quad \min_P s_i < 1 - R_i \quad (40)$$

also certify F . Indeed the affine functions $s_i - R_i$ and $1 - R_i - s_i$ are nonnegative on P and neither is identically zero. Each is positive throughout its relative interior: otherwise a small relative-neighborhood segment through a zero interior point would force an affine nonnegative function to vanish on the affine hull. Thus the response is strictly interior there, giving zero gradient.

The true box-QP response is continuous in x . More explicitly, adding its two variational inequalities at x, x' gives

$$\lambda_{\min}(Q) \|z(x) - z(x')\|_2 \leq \|C(x - x')\|_2$$

when the responses differ. Zero gradient therefore extends to all of P by continuity and density of its relative interior. Similarly, $\min_P t_i \geq S_i$ and $\max_P t_i > S_i$ certify L ; $\max_P t_i \leq -S_i$ and $\min_P t_i < -S_i$ certify U . These arguments apply in the affine hull of a lower-dimensional cell. The strict-somewhere conditions cannot be omitted.

Choose any sound available label for a certified coordinate. Call the others ambiguous, let t_P be their number on P , and put $t = \max_P t_P$. Stronger sound certificates may reduce this number. In particular, it measures the chosen certificates rather than an intrinsic complexity of the response.

Theorem 3.8 (Exact dense recovery after screening). *Suppose a valid surrogate cover has M nonempty rational polytopes in fixed dimension $r + s_0$, and whole-cell sound certificates leave at most t ambiguous coordinates on each. Exact bilevel feasibility and a rational global optimum, when feasible, can be obtained with at most $M3^t$ rational recovery LPs. Each LP and its preparation have polynomial bit complexity in the data and cover descriptions. This count excludes construction and screening of the cover. The displayed tests provide their certificates by rational matrix computations and vertex enumeration in fixed dimension, without additional LP calls for affine extrema. Starting with the supplied fixed-rank surrogate, the total bound is $3^t L^{O(r+s_0+1)}$, with L including all rational input and upper rows. No bound on the number of certified F coordinates is required.*

Proof. On a cover polytope P , keep all certified labels and enumerate the three labels for its ambiguous coordinates. A completed assignment partitions the coordinates into L, F, U . Set $z_L = 0$, $z_U = \mathbf{1}$, and solve the free equations with the true Hessian:

$$z_F(x) = -Q_{FF}^{-1}(c_F + C_F x + Q_{FU} \mathbf{1}). \quad (41)$$

Every principal submatrix of an SPD matrix is SPD, so it is invertible; the empty free set requires no inversion. Rational elimination has polynomial bit cost even when F is large and densely coupled. Impose

$$\begin{aligned} v \in P, \quad 0 \leq z_F(x) \leq 1, \\ (Qz(x) + c + Cx)_L \geq 0, \quad (Qz(x) + c + Cx)_U \leq 0, \end{aligned} \quad (42)$$

together with all original affine upper rows, and minimize the original upper objective after substitution. This is a rational LP in at most $r + s_0$ variables, bounded by compact P . Its feasible points satisfy all true box KKT conditions, so they give the unique response to the original dense follower.

Conversely, any true feasible leader–response pair has a surrogate graph point in at least one cell. Every certified label holds there. Assign an ambiguous coordinate label F when its gradient is zero, L when its gradient is positive, and U when its gradient is negative. Box optimality makes this a compatible assignment. Thus the pair is included in one recovery LP. Zero-gradient bound coordinates may appear under more than one assignment, which loses no solutions. The best nonempty LP yields an exact rational optimum; if all are empty, the original problem is infeasible. Summing 3^{t_P} over the cells gives the LP count. Fixed-dimensional cover construction and screening and polynomial rational inversion/substitution/LP give the stated bit bound. $\square$

An inexpensive additional sufficient test is often useful. Guess a full status assignment on P , form (41), and check all true KKT signs and bounds at every vertex of P . All tested functions are affine, so successful checks certify the response on the entire cell without status enumeration. Failure merely means that the assignment is not certified on the *whole* cell. It must remain available in the recovery enumeration on a smaller feasible part; a failed guess is not an infeasibility certificate.

3.5 A computable perturbation neighborhood

The ambiguity bound can be certified from the geometry of the supplied surrogate. Use its original closed clipping cells, retaining their nominal labels L, F, U . At a vertex v , call coordinate i a transition coordinate when

$$y_i(v) \in \{0, 1\}, \quad \hat{g}_i(v) = 0.$$

Let q be the largest number of such coordinates at a vertex of any nonempty cell. Vertices created by the leader polytope and consistency equalities are included; no general-position assumption is made.

Every such cell has affine dimension at most r . Its projection to x is injective, since a leader has a unique surrogate response and aggregate. An injective linear projection on a polytope is injective on its affine hull: a nonzero kernel direction in that affine hull would produce two nearby points with the same projection in a relative-interior neighborhood. Hence its dimension is at most the leader dimension.

Cover each cell by the convex hulls of all its affinely independent vertex subsets of size at most $r + 1$. These are rational simplices and they cover the cell by the finite-dimensional convex-combination reduction proved in lemma A.3. A triangulation is unnecessary: overlapping simplices are allowed. There are at most $\sum_{j=1}^{r+1} \binom{V_P}{j}$ such simplices for a cell with V_P vertices, polynomial in fixed dimension.

For a simplex T , let J_T be the union of its vertices' transition coordinates. Then $|J_T| \leq (r + 1)q$. If $i \notin J_T$, the containing cell's nominal label gives strictly positive vertex margins

$$L : \hat{g}_i(v), \quad U : -\hat{g}_i(v), \quad F : y_i(v) \text{ and } 1 - y_i(v).$$

For a nominal F coordinate its surrogate gradient vanishes throughout the closed cell, so any zero bound margin would indeed be a transition. Let $\sigma > 0$ be the minimum of all these margins over all simplices and coordinates outside their J_T , setting $\sigma = 1$ if the list is empty. It is rational, and every listed margin is at least σ throughout the corresponding simplex by affine interpolation.

For $N \geq 1$, compute

$$\begin{aligned} m_0 &= \frac{1}{\|\hat{Q}^{-1}\|_\infty}, \quad L_0 = \|\hat{Q}\|_\infty, \quad R = \lceil \sqrt{N} \rceil, \\ \varepsilon_0 &= \min \left\{ \frac{m_0}{2}, \frac{\sigma}{4R \max\{1/m_0, 1 + L_0/m_0\}} \right\}. \end{aligned} \tag{43}$$

Matrix infinity norms are induced maximum absolute row sums. Symmetry implies $\|A\|_2 \leq \|A\|_\infty$, so $\lambda_{\min}(\hat{Q}) \geq m_0$ and $\|\hat{Q}\|_2 \leq L_0$. The case $N = 0$ is an ordinary upper LP.

Corollary 3.9 (Transition-controlled dense neighborhood). *Every rational symmetric Q satisfying $\|Q - \hat{Q}\|_\infty \leq \varepsilon_0$ is positive definite. On each of the constructed simplices, at most $(r + 1)q$ coordinates require status enumeration in theorem 3.8. All radius and cover data can be computed with polynomial bit complexity for fixed r, s_0 , so exact optimization is polynomial for fixed r, s_0, q . The dense perturbation may have arbitrary rank and signs.*

Proof. Write $\varepsilon = \|E\|_\infty \leq m_0/2$. Then $\lambda_{\min}(Q) \geq m_0/2$. From (35), Cauchy–Schwarz and the unit-box bound $\|y\|_2 \leq R$ give

$$\|z - y\|_2 \leq 2\varepsilon R/m_0. \tag{44}$$

The true gradient minus the surrogate gradient equals $\hat{Q}(z - y) + Ez$. Since $\|z\|_2 \leq R$,

$$\|Qz + a - \hat{g}\|_2 \leq 2\varepsilon RL_0/m_0 + \varepsilon R \leq 2\varepsilon R(1 + L_0/m_0). \tag{45}$$

The choice of ε_0 makes both bounds at most $\sigma/2$. Every nontransition nominal L coordinate has strictly positive true gradient and hence is zero; a nominal U coordinate has strictly negative gradient and hence is one. A nontransition nominal F coordinate lies in $[\sigma/2, 1 - \sigma/2]$, so its true gradient is zero. These are sound whole-simplex certificates, independent of whether the ellipsoid tests happened to identify them. Only J_T needs enumeration.

Rational vertex enumeration, fixed-size simplex descriptions, matrix inversion, margin evaluation and the displayed norm calculations have polynomial bit cost. The simplex count above is polynomial for fixed r, s_0 ; it may increase the exponent of the original arrangement bound. Apply theorem 3.8 to this cover. $\square$

The radius is instance-dependent and can be small. Neither low transition multiplicity nor a numerically large margin is automatic. This is a verifiable neighborhood of a supplied surrogate, not a general method for finding a useful surrogate or a uniform neighborhood of every well-conditioned Hessian.

3.6 Value and feasibility bounds without full recovery

Even when many statuses remain ambiguous, the directional enclosure provides computable global bounds. For a signed output vector b , put

$$q_b(v) = b^\top y(v) - \frac{1}{2} b^\top Q^{-1} p(v), \quad r_b = \frac{1}{2} \sqrt{(b^\top Q^{-1} b) \eta_P}.$$

Choose a rational outward upper bound $\bar{r}_b \geq r_b$ by rational square-root enclosure. Its accuracy is an explicit computational choice; any such upper bound is safe. Then $q_b - \bar{r}_b \leq b^\top z \leq q_b + \bar{r}_b$ on P . For a row $a_j^\top x + b_j^\top z \leq h_j$, an inner sufficient condition and an outer necessary condition are, respectively,

$$a_j^\top x + q_{b_j}(v) + \bar{r}_{b_j} \leq h_j, \quad a_j^\top x + q_{b_j}(v) - \bar{r}_{b_j} \leq h_j. \quad (46)$$

Use these conditions with $v \in P$ in rational inner and outer LPs, representing equalities by both signs. If the objective is $a_0^\top x + b_0^\top z + c_0$, minimize $a_0^\top x + q_{b_0} - \bar{r}_{b_0} + c_0$ over the outer LPs. Their best value is a certified lower bound on the true optimum: every true feasible leader has a point in an outer LP and its objective is no smaller than that lower enclosure. Minimize the corresponding upper enclosure over the inner LPs for a certified upper bound and a leader feasible for the true response. Evaluating the true follower with an exact KKT certificate at that leader can improve this upper bound.

Empty outer LPs for every cell certify original infeasibility. Empty inner LPs alone say nothing about original feasibility, and a minimizer of an outer LP need not be truly feasible. The inner construction can miss exactly feasible upper equalities. Signed objective coefficients are handled by the directional formula, without a monotonicity assumption. Only the actual computed finite upper-minus-lower gap gives an objective guarantee; a small perturbation by itself supplies no small global loss under restrictive upper rows.

Example 3.10 (Small residual but complete screening ambiguity). Let $N \geq 2$, $\hat{Q} = I$, $Q = I + (\varepsilon/N)\mathbf{1}\mathbf{1}^\top$, $c = 0$, $C = -\mathbf{1}$, and $X = [0, 1]$, where $\varepsilon > 0$ is rational and arbitrarily small. The surrogate response is $y = x\mathbf{1}$ and the true response is $z = x\mathbf{1}/(1 + \varepsilon)$, whose gradient is zero throughout the interval. On the whole cell $[0, 1]$, however, the coordinate enclosure radius is positive and $s_i(0) = 0$, so neither strict nor refined F screening passes. The lower-gradient center has minimum zero and positive half-width, the upper-gradient test also fails, and the coordinate tests force neither bound. Thus the displayed ellipsoid tests leave all N coordinates ambiguous. Here $\|E\|_\infty = \varepsilon$ can tend to zero. The all- F guessed assignment in (41) immediately certifies the response

throughout the cell and needs no enumeration. This separates conservative screening from intrinsic difficulty.

Small perturbations can also move a switching point. With one follower, $\widehat{Q} = 1$, $Q = 1 + \varepsilon$, linear cost $-2xz$, and $x \in [0, 1]$, the upper clipping point moves from $1/2$ to $(1 + \varepsilon)/2$ for $0 < \varepsilon < 1$. Reusing the surrogate's upper-bound label on its entire cell would therefore be incorrect. Recovery must use the true KKT conditions. Dense rational elimination and construction of many cover cells can dominate cost even when screening is sound. The results do not establish practical superiority, a uniformly effective cell refinement, or automatic discovery of an accurate low-rank surrogate.

A Polyhedral blocks coupled through a fixed core

This appendix develops a related elimination principle for a single-level problem that is linear in many small polyhedral blocks and polynomial in a fixed-dimensional core. It both supplies a broader structural comparison and makes witness recovery explicit. Its convex combinations preserve polyhedral feasibility and linear measurements; they are not combinations of globally optimal responses of a nonconvex follower.

Fix $r, k \geq 0$ and $d \geq 1$. Let $C \subset \mathbb{R}^r$ be compact and given by an explicit rational quantifier-free polynomial formula. For $b = 1, \dots, B$, let

$$P_b(x) = \{y \in \mathbb{R}^{d_b} : G_b(x)y \leq h_b(x)\}, \quad 1 \leq d_b \leq d,$$

with polynomial rational data and explicit finite rational coordinate bounds among its rows. The local sets may be empty or lower-dimensional. Let $A_b(x) \in \mathbb{Q}[x]^{k \times d_b}$, $b(x) \in \mathbb{Q}[x]^k$, $c_b(x) \in \mathbb{Q}[x]^{d_b}$ and $c_0(x) \in \mathbb{Q}[x]$. Consider

$$\begin{aligned} \min_{x, z} \quad & c_0(x) + \sum_{b=1}^B c_b(x)^\top z_b, \\ \text{subject to} \quad & x \in C, \quad z_b \in P_b(x) \quad (b = 1, \dots, B), \\ & \sum_{b=1}^B A_b(x)z_b = b(x). \end{aligned} \tag{47}$$

Let L include all explicit data and let $\delta \geq 2$ bound actual input degree. In contrast with a fixed-degree restriction, the following statement tracks the numerical degree throughout its construction.

Theorem A.1 (Fixed-core polyhedral optimization). *For fixed r, d, k , feasibility and global optimization of (47) are computable in time polynomial in (L, δ) . If feasible, an optimum is attained and an optimizer with all block coordinates and its value can be returned in a single real-algebraic field with polynomial total encoding length. Neither the number of blocks nor their row counts is fixed. A fixed number of aggregate inequalities is also permitted if finite bounds for their slacks are supplied.*

A.1 Local vertices and support directions

For block b enumerate all d_b -row subsets I . Put

$$\Delta_{bI}(x) = \det(G_b(x))_I, \quad p_{bI}(x) = \text{adj}((G_b(x))_I)(h_b(x))_I.$$

Where $\Delta_{bI} \neq 0$, the candidate is p_{bI}/Δ_{bI} . Equivalently, use the positive-denominator form

$$D_{bI} = \Delta_{bI}^2, \quad Y_{bI} = \Delta_{bI} p_{bI}, \quad y_{bI} = Y_{bI}/D_{bI}. \quad (48)$$

Its validity predicate is

$$D_{bI} > 0, \quad G_b(x)Y_{bI}(x) \leq h_b(x)D_{bI}(x). \quad (49)$$

All these tests are polynomial of degree $O(d\delta)$.

Every vertex of $P_b(x)$ has d_b independent active rows. Otherwise there is a nonzero direction orthogonal to all active normals; a small step in either direction keeps the active rows equal and preserves all finitely many strict inactive rows, contradicting extremality. A nonempty bounded polyhedron has a vertex, and every linear functional has a maximizing vertex. For completeness, a maximizing face is nonempty and compact; successively minimize the coordinates within it to produce a singleton face, which is a vertex of the original polyhedron. These arguments also apply to lower-dimensional sets and singletons. Thus (48) lists enough candidates to detect nonemptiness and maximize every linear functional.

Append the objective as a measurement. With $h = k + 1$, define

$$W_b(x) = \begin{pmatrix} A_b(x) \\ c_b(x)^\top \end{pmatrix}, \quad T_b(x) = W_b(x)P_b(x), \quad t(x, \eta) = \begin{pmatrix} b(x) \\ \eta - c_0(x) \end{pmatrix}. \quad (50)$$

For a direction $\lambda \in \mathbb{R}^h$, the support score of a valid vertex is

$$\lambda^\top W_b(x)y_{bI}(x) = n_{bI}(x, \lambda)/D_{bI}(x), \quad n_{bI} = \lambda^\top W_b Y_{bI}.$$

Collect the denominator and validity polynomials, and, for every pair of candidates in one block, the comparison polynomial

$$n_{bI}D_{bJ} - n_{bJ}D_{bI}. \quad (51)$$

The denominators of valid candidates are positive, so these signs correctly order their scores. There are polynomially many polynomials for fixed d , all in the $r + h$ coordinates (x, λ) , of degree $O(d\delta)$ and polynomial coefficient bit length.

Enumerate their realizable sign conditions, remembering identically zero polynomials and including zero signs. Each sign condition determines the valid vertices and a support-maximizing vertex for every nonempty block; choose the first maximizer in a fixed ordering. Discard a condition if some block has no valid candidate. The retained list Σ is polynomial in size and can be constructed in polynomial bit time for fixed (r, d, k) . Disconnected sign realizations cause no problem, since both validity and score order are determined by signs.

A.2 Exact membership from the support inequalities

For $\sigma \in \Sigma$, let $I_b(\sigma)$ be its selected local index and abbreviate the corresponding denominator and numerator by $D_{b\sigma}, n_{b\sigma}$. Put

$$H_\sigma(x) = \prod_b D_{b\sigma}(x), \quad J_\sigma(x, \eta, \lambda) = \lambda^\top t(x, \eta)H_\sigma(x) - \sum_b n_{b\sigma}(x, \lambda) \prod_{a \neq b} D_{a\sigma}(x). \quad (52)$$

On that sign condition $H_\sigma > 0$ and $J_\sigma \leq 0$ states the support inequality

$$\lambda^\top t(x, \eta) \leq \sum_b \max_{y \in P_b(x)} \lambda^\top W_b(x)y.$$

Writing $S_\sigma(x, \lambda)$ for the sign conjunction, define

$$\Phi(x, \eta, \lambda) = \bigvee_{\sigma \in \Sigma} [S_\sigma(x, \lambda) \wedge J_\sigma(x, \eta, \lambda) \leq 0]. \quad (53)$$

Lemma A.2 (Support membership). *For $x \in C$, there exist blocks satisfying (47) with objective exactly η if and only if*

$$\forall \lambda \in \mathbb{R}^h \ \Phi(x, \eta, \lambda). \quad (54)$$

Proof. If a block is empty, no sign condition realized at (x, λ) is retained, for any λ . Thus (54) is false. Otherwise the Minkowski sum $T(x) = \sum_b T_b(x) = \{\sum_b t_b : t_b \in T_b(x)\}$ is compact and convex. Its support function is the sum of the support functions of its summands: a maximizing choice can be made independently in every block. The formula (53) therefore gives its exact support inequality in every direction.

A point t in a compact convex set T satisfies all its support inequalities. Conversely, if $t \notin T$, choose a nearest point $p \in T$ and put $\lambda = t - p \neq 0$. The derivative of squared distance along the segment from p to any $u \in T$ gives $(t - p)^\top(u - p) \leq 0$. Hence $\max_{u \in T} \lambda^\top u \leq \lambda^\top p < \lambda^\top t$, violating a support inequality. Thus (54) is equivalent to $t(x, \eta) \in T(x)$. By (50), this membership states precisely existence of blocks satisfying all linking equations and attaining the specified objective value. $\square$

The degrees in (52) are $O((B + 1)d\delta)$, which need not be constant. Their ambient dimension $r + h + 1$ is fixed, however, so their expanded monomial count and bit size are polynomial in (L, δ) by the product argument in lemma 2.3. Define

$$E(x, \eta) = (x \in C) \wedge \forall \lambda \ \Phi(x, \eta, \lambda).$$

This fixed-variable formula describes exact attainable core-value pairs. The full feasible set of (47) is closed and lies in the product of compact C and the finite block boxes. Therefore the upper minimum is attained whenever feasible. Apply quantifier elimination and joint sampling to

$$E(x, \eta) \wedge \neg \exists x', \eta' [E(x', \eta') \wedge \eta' < \eta]. \quad (55)$$

This gives an optimal (x^*, η^*) in one field $K = \mathbb{Q}(\alpha)$ of polynomial degree and encoding length. It remains to recover all block variables. Giving each of them a new quantified coordinate would destroy the fixed-dimensional argument.

A.3 Constructive recovery from a few support tuples

The same support enumeration already supplies a polynomial-size set of full block tuples sufficient for recovery. For each $\sigma \in \Sigma$ test its selected local denominators at x^* and discard the tuple if any is zero. Evaluate the remaining formulas and retain the tuple if every selected local vertex is feasible at x^* . Denote the retained tuples by $y^1, \dots, y^M$, and set

$$a^j = \sum_b W_b(x^*) y_b^j \in K^h \quad (j = 1, \dots, M).$$

There is no need to require that the original sign condition σ be realizable at x^* with some direction: feasibility of the tuple suffices for this filtering step. It never adds a point outside $T(x^*)$.

Lemma A.3 (A small common convex combination recovers every block). *The retained aggregate points satisfy $\text{conv}\{a^1, \dots, a^M\} = T(x^*)$. For every target $t \in K^h \cap T(x^*)$ one can find at most $h + 1$ retained tuples and nonnegative weights in K that sum to one and whose weighted aggregate is t , in polynomial bit time in the original input and target encoding. Using the same weights in every block gives a feasible original tuple with aggregate t .*

Proof. Every retained tuple is blockwise feasible, so its aggregate belongs to $T(x^*)$ and the indicated convex hull is contained in $T(x^*)$. For every real direction λ , the actual sign condition at (x^*, λ) is retained in Σ : no block is empty at an optimal core. Its selected vertices give a feasible tuple that attains the support of $T(x^*)$ in direction λ . This tuple survives the filter. Hence the convex hull has the same support function as $T(x^*)$. The separation argument in lemma A.2 proves equality.

The usual Carathéodory reduction gives a convex representation of a target in a finite convex hull with affinely independent support. To see this directly, start with a representation with all supported weights positive. If its lifted columns $(a^j, 1)$ are dependent, choose a nonzero dependence u . Its components sum to zero, so it has a positive component. Replace the weights θ_j by $\theta_j - su_j$, with $s = \min_{u_j > 0} \theta_j / u_j$. This preserves the target and the sum of weights, keeps weights nonnegative, and removes at least one support point. Repeat until the lifted columns are independent. Their number is at most $h + 1$.

For construction, enumerate all subsets of at most $h + 1$ retained points. On each affinely independent subset solve

$$\sum_j \theta_j a^j = t, \quad \sum_j \theta_j = 1.$$

Choose independent rows to solve this fixed-size system, then check all remaining equations and the nonnegativity of its unique weights. If a suitable subset exists over the reals, its independent linear system has its unique solution in K , since every coefficient and the target belong to K . Thus the enumeration finds it. There are at most $\sum_{q=1}^{h+1} \binom{M}{q}$ subsets, polynomial for fixed h . All arithmetic, zero tests, and comparisons take place in the already sampled ordered field K , whose degree and coefficient encodings are polynomial. Each system has fixed order, so determinant expressions and rational evaluation give polynomial coefficient lengths for the weights; summing the resulting fixed number of terms preserves the same bound for every recovered block coordinate.

Finally, put $z_b = \sum_j \theta_j y_b^j$ for every block. Convexity of $P_b(x^*)$ makes z_b feasible, and linearity of $W_b(x^*)$ gives $\sum_b W_b(x^*) z_b = t$. The same weights must be used in all blocks: this is how the prescribed total measurements are preserved. $\square$

Applying lemma A.3 to $t = t(x^*, \eta^*)$ recovers the original optimizer in the field K , completing the main claim of theorem A.1. With no blocks, the image sum is $\{0\}$ and the empty tuple is its sole support tuple; all the formulas use empty sums and products with their usual meanings. Blocks of dimension zero can be removed after testing their scalar rows. A fixed number of aggregate inequalities can be made equalities using bounded scalar slack blocks. Supplied slack bounds ensure that the result stays in the same compact model and do not change the fixed structural dimensions.

This argument gives polynomial dependence on the numerical input degree as well as input length: local determinants and pair comparisons have degree $O(d\delta)$; the global products have degree $O((B + 1)d\delta)$; sign enumeration, elimination and sampling are all in fixed dimension; and reconstruction adds only fixed-size algebraic linear systems. Under polynomially bounded degree, this is polynomial time in L .

The common support-direction construction has a direct predecessor in Gritzmann and Sturmfels [11, Algorithm 2.3.6, Theorem 2.3.7, and Corollary 2.3.10]. For fixed-dimensional input polytopes, they enumerate a shared arrangement of support directions and compatible maximizing summand

vertices, with polynomial binary complexity. Here the sign enumeration is uniform over the polynomially varying core; exact optimization and the subsequent reconstruction recover all original blocks in one common field. These additional steps are proved above. The support characterization, finite-dimensional convex-hull reduction, and real algebraic algorithms are established tools. An alternative recovery would solve a linear program over the sampled common field; Adler and Beling [1] analyze that route with explicit dependence on the common extension degree. The constructive recovery above uses the support tuples already generated and needs no large-dimensional algebraic LP solve. It depends on the affine block measurements, including the objective coordinate. It would not justify convexifying a nonconvex follower reaction set or mixing its tied responses to create additional follower choices.

References

- [1] Ilan Adler and Peter A. Beling. Polynomial algorithms for linear programming over the algebraic numbers. *Algorithmica*, 12:436–457, 1994. doi: 10.1007/BF01188714.
- [2] Daniel Arnström and Daniel Axehill. A unifying complexity certification framework for active-set methods for convex quadratic programming, 2020. URL <https://arxiv.org/abs/2003.07605v2>. arXiv:2003.07605, version 2.
- [3] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. On the combinatorial and algebraic complexity of quantifier elimination. *Journal of the ACM*, 43(6):1002–1045, 1996. doi: 10.1145/235809.235813.
- [4] Yasmine Beck, Ivana Ljubić, and Martin Schmidt. A survey on bilevel optimization under uncertainty. *European Journal of Operational Research*, 311(2):401–426, 2023. doi: 10.1016/j.ejor.2023.01.008.
- [5] Mathieu Besançon, Miguel F. Anjos, and Luce Brotcorne. Complexity of near-optimal robust versions of multilevel optimization problems. *Optimization Letters*, 15:2597–2610, 2021. doi: 10.1007/s11590-021-01754-9.
- [6] Mathieu Besançon, Miguel F. Anjos, and Luce Brotcorne. Robust bilevel optimization for near-optimal lower-level solutions. *Journal of Global Optimization*, 90:813–842, 2024. doi: 10.1007/s10898-024-01422-z.
- [7] Christoph Buchheim. Bilevel linear optimization belongs to NP and admits polynomial-size KKT-based reformulations. *Operations Research Letters*, 51(6):618–622, 2023. doi: 10.1016/j.orl.2023.10.006.
- [8] Cássio Fraga Dantas and Rémi Gribonval. Stable safe screening and structured dictionaries for faster ℓ_1 regularization. *IEEE Transactions on Signal Processing*, 67(14):3756–3769, 2019. doi: 10.1109/TSP.2019.2919404.
- [9] Xiaotie Deng. Complexity issues in bilevel linear programming. In *Multilevel Optimization: Algorithms and Applications*, pages 149–164. Springer, 1998. doi: 10.1007/978-1-4613-0307-7_6.
- [10] Michael R. Garey, David S. Johnson, and Larry Stockmeyer. Some simplified NP-complete graph problems. *Theoretical Computer Science*, 1(3):237–267, 1976. doi: 10.1016/0304-3975(76)90059-1.

- [11] Peter Gritzmann and Bernd Sturmfels. Minkowski addition of polytopes: Computational complexity and applications to Gröbner bases. *SIAM Journal on Discrete Mathematics*, 6(2): 246–269, 1993. doi: 10.1137/0406019.
- [12] Dorit S. Hochbaum and J. George Shanthikumar. Convex separable optimization is not much harder than linear optimization. *Journal of the ACM*, 37(4):843–862, 1990. doi: 10.1145/96559.96597.
- [13] Alan J. Hoffman. On approximate solutions of systems of linear inequalities. *Journal of Research of the National Bureau of Standards*, 49(4):263–265, 1952. doi: 10.6028/jres.049.027.
- [14] Sergey S. Ketkov and Oleg A. Prokopyev. On the complexity of bilevel linear and quadratic programs in fixed dimensions, 2026. URL <https://arxiv.org/abs/2511.15592v2>. arXiv:2511.15592, version 2, June 10.
- [15] Thomas Kleinert, Martine Labbé, Ivana Ljubić, and Martin Schmidt. A survey on mixed-integer programming techniques in bilevel optimization. *EURO Journal on Computational Optimization*, 9:100007, 2021. doi: 10.1016/j.ejco.2021.100007.
- [16] Jun Liu, Zheng Zhao, Jie Wang, and Jieping Ye. Safe screening with variational inequalities and its application to Lasso. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *Proceedings of Machine Learning Research*, pages 289–297, 2014. URL <https://proceedings.mlr.press/v32/liuc14.html>.
- [17] Yi-Hsin Liu and Thomas H. Spencer. Solving a bilevel linear program when the inner decision maker controls few variables. *European Journal of Operational Research*, 81(3):644–651, 1995. doi: 10.1016/0377-2217(94)00005-W.
- [18] Nimrod Megiddo and Arie Tamir. Linear time algorithms for some separable quadratic programming problems. *Operations Research Letters*, 13(4):203–211, 1993. doi: 10.1016/0167-6377(93)90041-E.
- [19] Nagisa Sugishita and Margarida Carvalho. Complexity of bilevel linear programming with a single upper-level variable, 2026. URL <https://arxiv.org/abs/2510.21126v2>. arXiv:2510.21126, version 2, March 20.
- [20] Antoine Vigneron. Geometric optimization and sums of algebraic functions. *ACM Transactions on Algorithms*, 10(1), 2014. doi: 10.1145/2532647.